A Plot-Level Hazard and Living Suitability Screen for Hill Land

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Abstract

National and state-level hazard maps are built to guide planning decisions at the scale of a district or a watershed. They have no demonstrated skill at the scale that matters to a buyer, lender, or developer: one plot of land. This paper describes a screening model that scores individual hill-land parcels in Uttarakhand and Himachal Pradesh, India, for landslide, flood, and glacial-lake-outburst hazard, and for living suitability, at 30-metre ground resolution inside each of 136,935 scored patches. The model is built from satellite elevation, open hazard and climate records, and road-network access data, scored against fixed, pre-declared thresholds. It is tested against landslide, flood, and outburst events the model had not seen, and separately against live property-listing prices, a test the model was not built to pass. Results, data sources, and stated limits follow.

1. Introduction

Hill land plots that look alike on a brochure behave very differently on the ground. Two parcels a few hundred metres apart can sit on stable terrain and on an active slide path respectively, differ by tens of metres in distance from a seasonal torrent, or fall on opposite sides of a glacial-lake-outburst corridor — none of which shows up in a listing photo or a plot boundary survey.

Hazard information that does exist is built for a different scale. National landslide susceptibility maps and state disaster atlases are produced to guide district- or watershed-level planning. Tested directly against documented landslide events recorded after the model was built, a national susceptibility layer of this kind showed no usable skill at the scale of an individual parcel. The gap is not a flaw in those maps — they were not built for plot-level decisions — but it leaves buyers, lenders, developers, and the lawyers and brokers who advise them without a number that applies to the specific plot in front of them.

This paper describes a model built to close that gap: one score, and a set of underlying measurements, per plot, each traceable to a named source, each tested against events the model did not see during development.

2. Coverage

The screen covers, starting with Uttarakhand and Himachal Pradesh:

- 136,935 scored patches, each approximately 0.8 km² — an H3 hexagonal cell at resolution 8.
- 2 states, 20 districts, 200 tehsils (sub-districts).
- 30-metre ground resolution for terrain measurements inside each patch.
- 80 or more measured values, plus 27 climate and hazard rows (each given a fixed 1–5 score), per plot report, organized across 9 report sections.

The report structure was cross-checked line by line against two published international sample report editions for hill land. All 72 line items of those editions are accounted for in this product: each has either a matching data source and licence, or an explicit stated reason it is not, or not yet, included.

3. Method

The score for each patch is built from four layers of measurement. The exact thresholds, factor weights, and internal rule structure used to combine them are not published (Section 7); what follows describes what is measured and how it is checked, not the rules themselves.

3.1 Terrain geometry

Slope, elevation, aspect, and drainage are derived from satellite elevation data at 30-metre resolution. A row/column duplication artifact was found in the raw elevation export during development — a resampling effect of the export process, not a feature of the terrain — and corrected before any terrain-derived layer was computed from it.

3.2 Hazard exposure

An independent landslide-susceptibility estimate, river and torrent proximity, and traced glacial-lake and glacial-lake-outburst-flood corridors are computed for every patch. Where two or more of these layers cover the same ground, they are cross-checked against each other.

3.3 Climate and access

Heat, frost, fire weather, drought, heavy rainfall, and wind are drawn from open climate records. Access is measured as road-network travel distance — not straight-line distance — to the district capital, the nearest highway, the nearest hospital, the nearest school, and the nearest market town.

3.4 Fixed, pre-declared thresholds

Every hazard band and flag is set against a physical cutoff decided before scoring begins. None of these cutoffs are rescaled relative to whatever else happens to be on the map on a given run. The practical consequence: the same ground reads the same way regardless of what else is nearby or what else has been scored that day.

The remainder of this paper — validation results, market and quality-control checks, limitations and data sources — is available on request.
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events, each against matched ground nearby — the same state block, the same terrain class, the same elevation band, and in some comparisons the same road-distance band.

Table 2. Matched-ground backtest results

Flag or band	Comparison	Lift over matched ground
Terrain + river flood flag	National landslide catalogue	[illegible]
Terrain + river flood flag	Documented flood incidents	[illegible]
Glacial-lake outburst corridor	[illegible] documented outburst events, [illegible] base rate on matched ground	[illegible]

The observed maximum flood grade was checked for accuracy (AUC) against an independent satellite-derived flood map: [illegible] overall, and [illegible] for locations with more than one recorded flood.

4.3 Event-fitted scores against [illegible] set-aside events

The combined hazard score is a model fitted to recorded events. Its landslide and flood terms were fitted (shallow gradient-boosted trees, fixed settings) on [illegible] recorded landslide, flood and outburst events with exact or village-level locations, using terrain, channels and height above them, rainfall, soil, vegetation and built-up ground; the flood term is averaged half and half with the physical flood score, and the earthquake term is unchanged. Another [illegible] events ([illegible]) were set aside before fitting, split by whole map cells so that neighbouring events stay on one side, and balanced by state block and event type. Every design choice was made on the [illegible] events, which were inspected over three rounds of changes; the set-aside events were scored once for the decision. A robustness check run afterwards (comparison ground inside the set-aside cells, which the model had not seen in training) gave the figures in Table 3; with comparison ground the model had seen, the first read gave an AUC of [illegible]. Each set-aside event was compared with ground of the same type in the same region that had no records: same state block, terrain class, river setting and road-distance band, using comparison ground the model had not seen.

Table 3. Event-fitted score against matched ground, [illegible] set-aside events

Measure	Result
Accuracy (AUC), [illegible] chance; [illegible] of the [illegible] events matched to comparison ground	[illegible] to [illegible] cells, resampled by map cell, no refits

[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

5. Market validation

[illegible]

- Across 300 localities (1,029 listings), the suitability score (shown in reports as Living Suitability, with no hazard deduction) tracks relative price: Spearman $\rho = +0.22$ (95% CI +0.16 to +0.31).
- On hill terrain (293 localities) the link is stronger: $\rho = +0.39$ (95% CI +0.26 to +0.48); On plains and foothills there is no clear link ($\rho = +0.05$, 95% CI −0.04 to +0.15). Outside Dehradun, hill localities give $\rho = +0.26$ (95% CI +0.10 to +0.40), so the hill result depends on Dehradun. Using the largest portal alone (226 localities, 130 on hills), hill $\rho = +0.39$ (95% CI +0.26 to +0.51) and outside Dehradun +0.28 (95% CI +0.04 to +0.47). Widening the neighbourhood to two, three or five rings of hexagons does not change this.
- The physical total hazard leans towards a discount on hill land ($\rho = -0.15$, 95% CI −0.26 to −0.03), and so does the physical landslide driver ($\rho = -0.17$, 95% CI −0.31 to −0.02); the earthquake driver follows region, not price. The crowd-filtered score shows no price discount ($\rho = +0.04$, 95% CI −0.04 to +0.13), a likely reason is that settled towns have both higher prices and more recorded events (not tested).
- In 40 Himachal localities (mostly a single listing each) neither suitability ($\rho = -0.02$) nor physical total hazard ($\rho = -0.05$, 95% CI −0.40 to +0.30) shows a clear link with price; the sample is too small to say more.
- As a check on the listing data itself, plot rates published by a property portal for 25 Dehradun localities (price per square yard, plots only; apartment rates were kept apart) agree with the median of our own listings for the same localities: $\rho = 0.86$.

Earlier figures from a smaller sample (48 localities) were stronger and did not hold when the sample grew; the figures above replace them.

6. Quality control

Every report is checked before it is issued, and the scoring is checked before it is sent.

- 3,186 of 3,186 automated checks pass across 70 test plots covering every report section, with zero failures.
- All scored patches reproduce bit-for-bit on an independent re-run of the scoring pipeline — a given plot's score does not depend on when or how it was requested.
- A targeted quality pass drew a 10,000-patch sample from a 15 km radius around 15 named towns with active property interest (including Mussoorie, Dehradun, Nainital, Shimla, Manali and Dharamshala). Automated geometric and statistical consistency checks were run — recomputing terrain class and the final score directly from stored inputs and diffing against the stored result, checking hazard flags against physical inputs, ranking patches by statistical deviation from same-terrain, same-hazard-band peers. The 500 most statistically unusual patches, plus a separate 100-patch random control group,

were then independently reviewed for geographic plausibility. Zero confirmed anomalies in either group.

7. Limitations

- This is a relative-risk screening tool, built on a model tested against historical disaster records and compared with asking prices. It does not replace official certifications or the qualified engineering, legal, and administrative requirements a purchase, loan, or build still needs.
- The landslide event samples are modest in size ([illegible], [illegible], [illegible], [illegible] and [illegible] events respectively). The lift and accuracy figures in Section [illegible] should be read as direction, not as an exact probability for any one plot.
- The event-fitted score uses built-up ground, so part of its ranking reflects where events get recorded (settled ground and roads). On the [illegible] verifying events, matching comparison ground on built-up share kept more than half of the gain over the physical score, not all of it. The landslide term also takes a published susceptibility map as one input, and that map's training events may overlap the test events. The test covers so many cells; [illegible] of the [illegible] events were on hill or high-mountain terrain and [illegible] on plains, so the event-fitted score is reported for sloping hill land only and the physical drivers stand alone on flat ground: plains and foothills. The Geological Survey of India mapped slides in Himachal improve least. The earlier version [illegible] landslide driver without the susceptibility map [illegible] built-up share alone [illegible] without matching) describe the earlier physical score.
- Event records skew toward locations near roads and towns — a known reporting bias in disaster records generally.
- The market check in Section [illegible] uses asking prices, not sale prices. Over half of the localities have a single listing, and Uttarakhand supplies many of the hill localities; without it the hill link is weaker ([illegible] interval includes zero). The score is a ranking, not a probability.
- The exact scoring thresholds, factor weights, and internal rule set are not published. Method, full data sources, and complete test results are available on request.

8. Conclusion

The model scores [illegible] patches across Uttarakhand and Himachal Pradesh for hazard and living suitability, at [illegible]-metre ground resolution, from measurements that trace to named open sources. It separates events it had not seen in development from similar nearby ground at rates above chance, as reported in Section [illegible], and its living suitability score shows a modest link with asking prices on hill land, mostly through Uttarakhand, as reported in Section [illegible]. Both checks are reported with their sample sizes and confidence intervals, and the limits of each are stated in Section 7.

9. Data sources

- Copernicus DEM (elevation)
- OpenStreetMap contributors (ODbL) — road network and access
- ERA5-Land — climate records
- JRC Global Surface Water
- ICIMOD glacial lakes
- Randolph Glacier Inventory
- Geological Survey of India landslide inventory
- WorldPop